

RayEncoder Evaluation Report

Checkpoint: checkpoint/001_RayEncoder/checkpoint/ray_encoder_best.pth

Split: val | Samples: 1200

8-way direction accuracy: 47.33%

Ray cosine: mean 0.8739, median 0.9348, p10 0.6568, p90 0.9920

Angular error: mean 34.72 deg, median 24.06 deg, p90 83.48 deg

Temperature error: mean 0.1718, median 0.1808

Confidence: mean 0.4889, median 0.4892

Confusion Matrix

Rows are ground truth, columns are predicted. Diagonal cells are correct.

		Predicted direction							
		N	NE	E	SE	S	SW	W	NW
Ground Truth	N	83 55%	36 24%	3 2%	0 0%	6 4%	4 3%	3 2%	15 10%
	NE	15 10%	89 59%	26 17%	5 3%	5 3%	5 3%	0 0%	5 3%
	E	4 3%	9 6%	94 63%	27 18%	1 1%	5 3%	5 3%	5 3%
	SE	0 0%	0 0%	35 23%	80 53%	20 13%	5 3%	10 7%	0 0%
	S	0 0%	0 0%	5 3%	48 32%	70 47%	17 11%	10 7%	0 0%
	SW	1 1%	0 0%	1 1%	18 12%	51 34%	41 27%	33 22%	5 3%
	W	7 5%	3 2%	0 0%	0 0%	19 13%	32 21%	63 42%	26 17%
	NW	39 26%	10 7%	6 4%	1 1%	6 4%	6 4%	34 23%	48 32%

Best 5 Samples by Angular Error



Image013_SW_3500.png

scene: Image013
GT: SW (225 deg)
Pred: SW (225.0 deg)
angle error: 0.0 deg
ray cosine: 0.9995
temp pred/gt: -0.824 / -0.800
confidence: 0.490



Image013_NE_2500.png

scene: Image013
GT: NE (45 deg)
Pred: NE (44.9 deg)
angle error: 0.1 deg
ray cosine: 0.9501
temp pred/gt: -0.978 / -1.200
confidence: 0.494

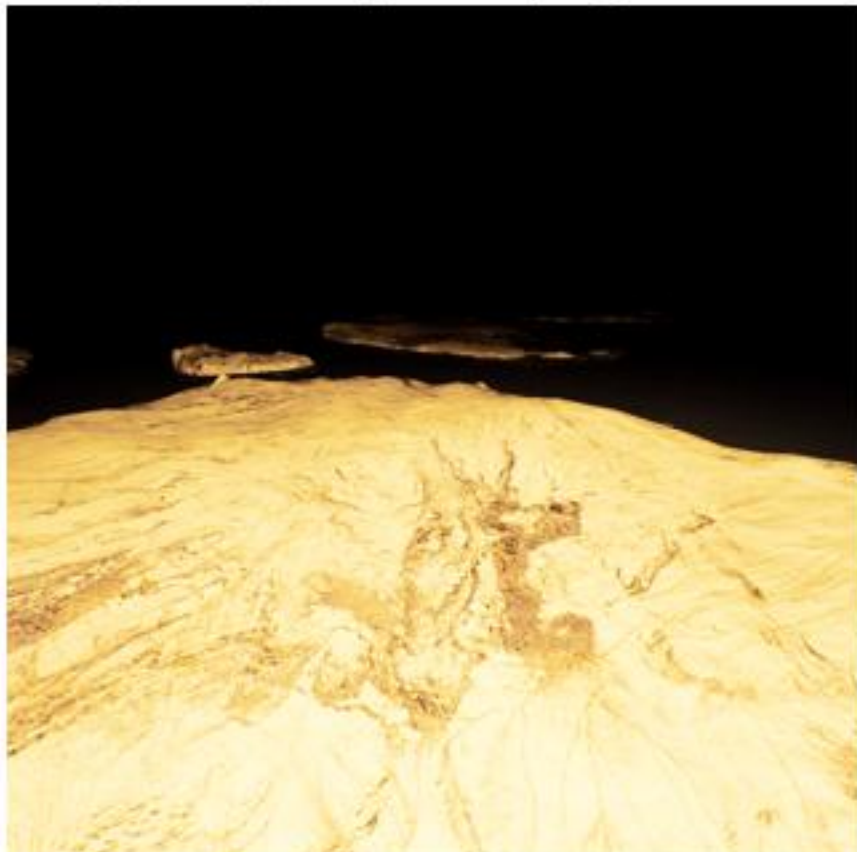


Image079_S_3500.png

scene: Image079
GT: S (180 deg)
Pred: S (180.2 deg)
angle error: 0.2 deg
ray cosine: 0.9998
temp pred/gt: -0.760 / -0.800
confidence: 0.481



Image143_W_6500.png

scene: Image143
GT: W (270 deg)
Pred: W (270.1 deg)
angle error: 0.1 deg
ray cosine: 0.9824
temp pred/gt: 0.601 / 0.400
confidence: 0.486



Image184_N_5500.png

scene: Image184
GT: N (0 deg)
Pred: N (359.9 deg)
angle error: 0.1 deg
ray cosine: 0.9942
temp pred/gt: 0.014 / 0.000
confidence: 0.493

Worst 5 Samples by Angular Error



Image003_NE_5500.png

scene: Image003
GT: NE (45 deg)
Pred: SW (225.1 deg)
angle error: 179.9 deg
ray cosine: 0.5433
temp pred/gt: -0.139 / 0.000
confidence: 0.491



Image003_N_3500.png

scene: Image003
GT: N (0 deg)
Pred: S (182.1 deg)
angle error: 177.9 deg
ray cosine: 0.3584
temp pred/gt: -0.855 / -0.800
confidence: 0.492



Image088_E_3500.png

scene: Image088
GT: E (90 deg)
Pred: W (267.6 deg)
angle error: 177.6 deg
ray cosine: 0.3692
temp pred/gt: -0.926 / -0.800
confidence: 0.488

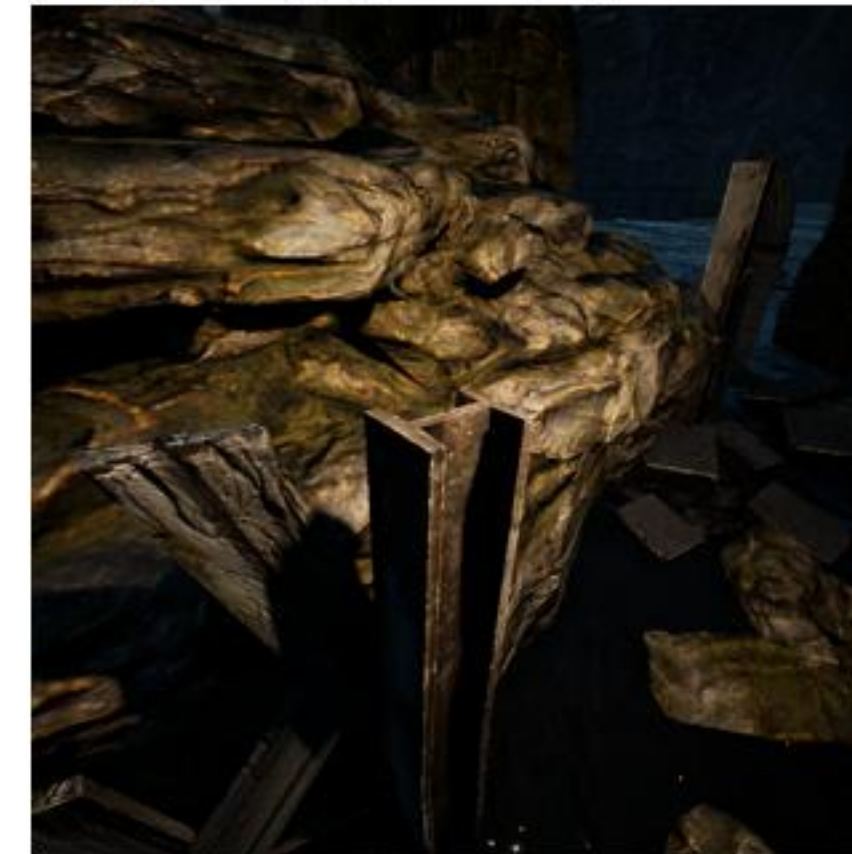


Image088_E_4500.png

scene: Image088
GT: E (90 deg)
Pred: W (272.8 deg)
angle error: 177.2 deg
ray cosine: 0.3957
temp pred/gt: -0.652 / -0.400
confidence: 0.488

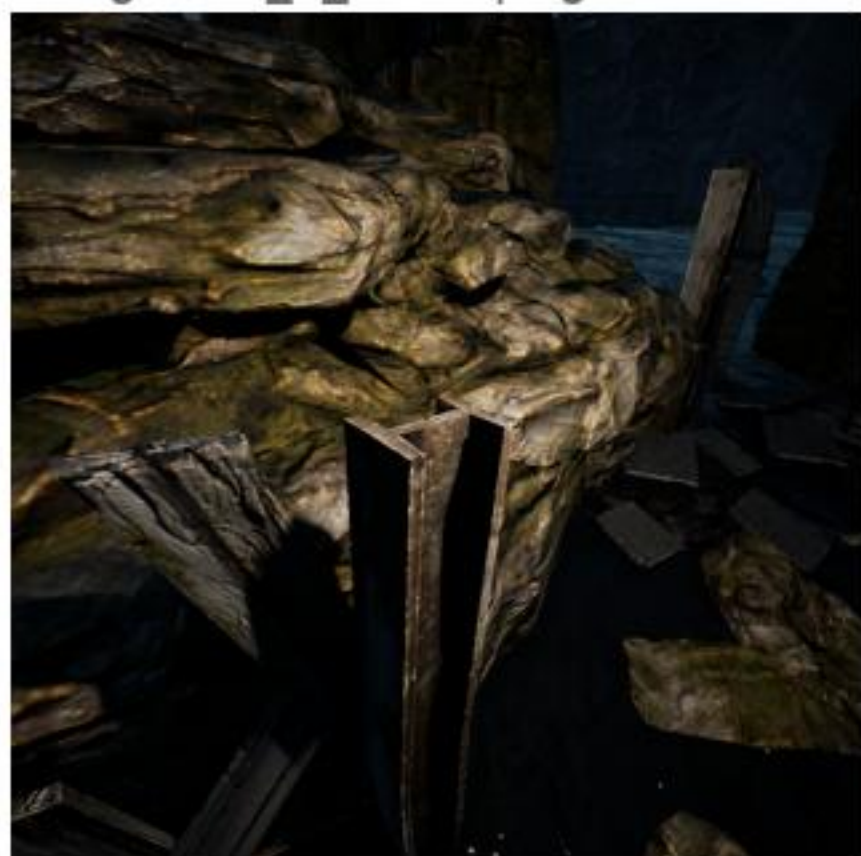


Image088_E_5500.png

scene: Image088
GT: E (90 deg)
Pred: W (273.3 deg)
angle error: 176.7 deg
ray cosine: 0.4228
temp pred/gt: -0.220 / 0.000
confidence: 0.487